[PZoom Deep Dive 9.2] Verify that β is a transitive set

To show that β is transitive, we assume $x \in y$ and $y \in \beta$. We must deduce that $x \in \beta$.

Because β is an element of the ordinal α, and α is a transitive set, every element of α is also a subset of α. Thus, $\beta \in \alpha$ implies $\beta \subseteq \alpha$. Since $y \in \beta$, it follows that $y \in \alpha$.

Applying transitivity to α once more, since $y \in \alpha$, we have $y \subseteq \alpha$. Because $x \in y$, it follows that $x \in \alpha$.

We now have three elements (x, y, and β) all belonging to the ordinal α. On α, the membership relation $\in$ serves as a strict total ordering. The statements $x \in y$ and $y \in \beta$ are definitionally equivalent to $x < y$ and $y < \beta$. By the transitivity of the strict total order $<$ on α, we conclude that $x < \beta$, which translates back to:

$$x \in \beta.$$

Since x was an arbitrary element of y, this proves $y \subseteq \beta$. Thus, β is a transitive set.

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